

ChemEAGLE: cost and quality comparison

Decision report | 1 October 2026 | HKUST API | 73 images / 893 ground-truth reactions

Recommendation

Use Gemini 3.7 Flash with default reasoning and selective GPT-5.6 Sol high review. The optimized variant preserves benchmark quality while substantially reducing estimated token cost and observed latency. The cheapest open-weight candidates were much less accurate when asked to convert chemistry images directly into SMILES.

Comparison with current ChemEagle

The reference is the saved ChemEagle Sol medium extraction plus Sol high review run from 21 September. It is the strongest saved two-pass baseline evaluated here. It was rescored with the same current evaluator as the new runs; the original pipeline was not rerun live. A separate upgraded local checkout already defaults to Gemini 3.7 Flash, so these findings compare a specific recorded pipeline, not every ChemEagle configuration.

Metric	Saved ChemEagle baseline	Cost-optimized ChemEagle	Change
Benchmark F1	0.8150	0.8199	+0.0048
Exact F1	0.8038	0.8176	+0.0138
Successful images	73/73	73/73	Same
Mean seconds per image	157.8	39.4	75.0% lower
Reference token cost per 73 images	USD 9.159	USD 1.926	79.0% lower
Reference token cost per image	USD 0.1255	USD 0.0264	Same pricing basis
Model calls	146	78	46.6% fewer
Invalid SMILES	6	0	Syntax validity improved

The small F1 improvement is not proof of superiority: this is a single benchmark run, and the review policy was developed on a 12-image subset reused in the full corpus. Treat the result as evidence of comparable quality at lower cost. F1 is a precision/recall measure, not the percentage of completely correct images.

At the same measured usage and reference rates, 1,000 similar images would cost approximately USD 125.47 for the baseline versus USD 26.39 for the optimized variant, saving USD 99.08. This is a linear illustration, not an HKUST quote; image complexity, retries and review frequency can change it.

Full benchmark

Configuration	Successful / images	Benchmark F1	Exact F1	Mean seconds/image	Reference USD
Saved Sol medium + high review	73/73	0.815	0.804	157.8	\$9.159
Lite: Gemini Flash + selective Sol review	73/73	0.820	0.818	39.4	\$1.926
Gemini 3.7 Flash, default reasoning	73/73	0.795	0.766	32.4	\$1.466
Gemini 3.7 Flash, low reasoning	73/73	0.777	0.692	24.0	\$0.934
Kimi K3, reasoning off	73/73	0.265	0.229	27.2	\$0.888
Qwen 3.8 Flash service, reasoning off	72/73	0.169	0.132	19.3	\$0.074
MiniMax M3, reasoning off	70/73	0.113	0.055	49.5	\$0.130

All seven configurations above cover the same 73 available images and 893 ground-truth reactions. Every image was attempted in the live full runs. Failed responses contribute zero predictions while keeping their ground truth in the recall denominator. Qwen had one final failed response; MiniMax had three. Successful results were reused when extending probes into full runs. Token costs include retained billed retries with reported usage.

Why this saves cost

The baseline calls Sol twice for every figure, including a full image and three overlapping crops for the review. Lite uses one Flash call per figure and local RDKit checks for invalid SMILES, unresolved wildcard atoms, empty results and duplicate reactions. Only **5 of 73 images** triggered the expensive visual review: **78 total model calls**, versus 146 baseline calls. No ground truth, expected row counts or expected molecules were sent to either model.

All five reviews preserved row count and row identities, and none increased invalid SMILES, unresolved placeholders or duplicates. The reviews corrected all 68 invalid source SMILES; the final result had **0 invalid SMILES among 2,543 molecular entries**. One image retains a duplicate-row review flag. Valid SMILES alone do not prove that a graph agrees with its drawing.

Clean results can be cached by image, prompt, model, reasoning, resolution and review policy. The live cache smoke test returned the same result with one paid request on the first invocation and zero requests on the second. Lite runs on CPU without specialist checkpoints or PyTorch/CUDA.

Other settings tested

Configuration	Successful / images	Benchmark F1	Exact F1	Mean seconds/image	Reference USD
Gemini Pro 3.1, low reasoning, 16k cap	12/12	0.452	0.382	12.1	\$0.213
Gemini Pro 3.1, default reasoning, 16k cap	5/12	0.194	0.194	106.2	\$3.114
Gemini Pro 3.1, default reasoning, 32k cap	11/12	0.456	0.456	145.6	\$2.890
Kimi K3, low reasoning, 16k cap	8/12	0.302	0.302	186.4	\$0.448

These rows use the same separate 12-image probe, covering scopes with 3-33 reactions, and its 156 ground-truth reactions. Gemini Pro's 16k budget was often consumed by reasoning, producing truncated or invalid JSON. Raising its budget to 32k improved completion reliability but remained expensive and weaker on this task. Unknown usage from earlier timeouts makes those prices lower bounds. MiniMax's default-reasoning experiment completed only one of eight attempts; seven timed out and four other probe inputs were left unattempted after the account-wide spend checkpoint. It is excluded from the quality ranking.

MiniMax M3 and Kimi K3 are the open-weight candidates. Their tested direct image-to-SMILES routes lost substantial accuracy. The Qwen Flash service was also tested; its weight availability was not verified. A future experiment could combine open models with specialist structure recognition rather than asking them to reconstruct every SMILES directly.

Cost and measurement limits

USD figures use the saved 1 October [OpenRouter model catalog](#) as common reference pricing, including reported reasoning tokens and cache discounts. They are estimates, and HKUST's actual routing rates can differ. Provider-reported costs, token details and unknown-usage counts are retained in the JSON score reports. The observed HKUST account balance fell from HK\$277.84 to HK\$168.92, a **HK\$108.92 account-wide delta** across all probes, retries, full runs and the cache smoke test. Concurrent outside account activity and late provider billing can affect that observation. No credits were purchased.

The baseline consists of saved 21 September predictions, rescored today using the same evaluator and current RDKit. Its latency includes both extraction and review; its cost re-prices both stages at the common reference rates. Live runs used bounded concurrency, so latency is an observed request measurement under that workload rather than a controlled infrastructure comparison.

GT4 contains 78 rows but only 73 corresponding PNGs are available locally. Only the existing colon-to-underscore filename mismatch was normalized. The new images were bounded to 1,800 pixels on the longest side. Soft F1 uses the existing 0.5 reactant/product overlap rule. Exact F1 requires complete matching reactant/product multisets. Both canonicalize SMILES, strip stereochemistry and keep the largest salt fragment. Conditions and PDF extraction are outside this experiment. These results do not establish performance on other datasets.

The upgraded local ChemEagle checkout already defaults to `gemini-3.7-flash`, which informed that candidate. Yufan's email has **not been verified**: the HKUST mailbox requires user sign-in in the in-app browser. No email was sent.

Implementation and reproduction

Branch: `mathus/chemeagle-cost-optimized`. The original dirty checkout was preserved; this work is isolated in `ChemEagle-efficient`.

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.efficient.txt
# Set HKUST_API_KEY privately.
.venv/bin/python efficient.py figure.png --output result.json --cache .cache --review
```

Omit `--review` for the measured single-pass mode. Reduced reasoning is available with `--reasoning economical`, with the quality tradeoff shown above. Failed, truncated or locally flagged results are not cached. The original multi-agent pipeline is retained. Ten focused tests passed, and all five accepted live reviews passed the final deterministic gates.

The score JSONs, `comparison.csv`, runtime versions, dataset manifest and per-image prediction artifacts accompany this report. `benchmarks/run_efficient.py` repeats the API runs; `benchmarks/report_efficient.py` repeats the scoring; `benchmarks/review_flagged.py` repeats selective review.

API provenance and model availability

All live inference requests in this experiment used the HKUST API, at `https://hkust.azure-api.net/hkust-genai/v1/chat/completions`. OpenRouter was used only as a public reference pricing catalog; the experiment did not send inference requests directly to OpenRouter.

The Sol baseline and selective reviewer use `gpt-5.6-sol`. The HKUST model catalog checked on 1 October did not list GPT-6 Sol or GPT-6.1 Sol. They were not benchmarked, and selecting either in Codex or signing into ChatGPT does not demonstrate access through HKUST. No conclusion about their chemistry quality or cost follows from this study.

Deployment decision

Keep the optimized variant opt-in while validating it on an independent held-out dataset, particularly stereochemistry and salts. Use the Gemini default-reasoning single pass only when its modest quality reduction is acceptable. Do not replace the baseline with the tested Kimi, Qwen or MiniMax direct-conversion configurations on the basis of their low price alone.

The original pipeline remains available on this branch. Clean extraction results can be cached; live smoke verification showed zero API calls on the second identical request. Failed, truncated or flagged results do not enter the clean cache. The branch is mathus/chemeagle-cost-optimized in [Tew12345678910/ChemEagle](#).

Evidence files

- `reports/comparison.csv`: full-corpus comparison, including invalid SMILES counts.
- `reports/selective-review.json`: optimized profile scores and usage.
- `reports/predictions/`: per-image predictions for independent inspection.
- `reports/dataset_manifest.json`: benchmark image inventory and hashes.
- `reports/hkust_model_catalog_2026-10-01.json`: available HKUST models.
- `reports/openrouter_catalog_2026-10-01.json`: common pricing snapshot.
- `reports/account_balance.json` and `reports/cache_smoke.json`: account observation and live cache check.

The Markdown and PDF versions contain the same report. Ten focused implementation tests passed in the completed benchmark work; no additional inference spend was needed to prepare this report.